

1. Given an integer array, find the length of the longest contiguous segment where:
The frequency of the most frequent element
is strictly greater than the sum of frequencies of all other elements in that segment.

Input 1: [1,2,1,1,3,1,2]

Output 1: Output: 4

Input 2: [5]

Output 2: 1

Input 3: [1,2,3,4]

Output 3: 1

2. Given a string, repeatedly delete any substring of length ≥ 2 consisting of the same character.
Return the final string after no more deletions are possible.

Input 1: "abbbaacc"

Output 1: ""

Input 2: "abcd"

Output 2: "abcd"

Input 3: "abccdb"

Output 3: "abdb"

3. Given an array, find the minimum distance between two equal elements.
Return -1 if none.

Input 1: [7,1,3,4,1,7]

Output 1: 3

Input 2: [1,2,3,4]

Output 2: -1

Input 3: [5,5,5,5]

Output 3: 1

4. Given a string s, rearrange its characters such that:
No two adjacent characters are the same.
If impossible $\rightarrow$ return "".

Input 1: "aab"

Output 1: "aba"

Input 2: "aaab"

Output 2: ""

Input 3: "aaabc"

Output 3: "abaca"

5. You are given a string s consisting of lowercase and uppercase English letters.
Return the longest contiguous substring of s that is a palindrome.
If multiple answers exist, return any one of them.

Input 1: "babad"

Output 1: "aba"
Here, also "bab" possible.

Input 2: "cbba"
Output 2: "bb"

Input 3: "a"
Output 3: "a"